Electrocatalytic Water Disinfection Using Black Titania

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Abstract

Biological pathogens in our water supplies are becoming increasingly resistant to commercial disinfectants. Advanced oxidation processes (AOPs) are being explored as a means to provide more powerful radical species to combat these more challenging contaminants. One commonly studied AOP for waterborne pathogen disinfection is TiO_2-catalyzed UV oxidation, which leads to the direct production of $OH^{\bullet}$ at the TiO_2 surface. The disinfecting power of this AOP can be enhanced by introducing defects into the TiO_2 band structure during annealing in a reducing atmosphere, creating 'black titania.' Recently, our group used black titania nanotubes as electrocatalysts, producing aqueous radicals under only an applied voltage (no UV radiation). UV-visible spectroscopy and x-ray photoelectron spectroscopy revealed the presence of Ti^{3+} and Ti-OH groups, both of which contribute to the continuous electronic exchange with species in the electrolyte solution and the production of an oxidizing titania surface under an anodic bias. The electrocatalytic disinfection device, consisting of a black titania anode and a stainless steel cathode, provides 6 log removal of E. *coli* from a starting influent concentration of 10^7 CFU/100 ml in 15 s.

Biography

Krista Carlson received her PhD in Glass Science from Alfred University in 2008. After graduation, she worked at Corning, Inc. where as a development scientist, she became the lead on two patents involving the chemical treatment of glass to form anti-glare surface properties. In 2016, she joined the Department of Metallurgical Engineering at the University of Utah. Her research program is centered on the development of innovative materials-based solutions to anthropogenic related environmental issues. Within this environmental theme, her research platform focuses on the development of materials and applications in the following areas: 1) customized aerogel membranes, 2) integrated waste containment and glass recycling technologies, and 3) advanced materials for decentralized drinking water purification technologies.



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